

might not be successful. Make sure to include power supply details, signal sources, component models with component labels in accordance with the rule file, and the same for net labels, too. One advantage is that gschem lets you include models either by setting the component's file attribute or by using a SPICE Model object.

Compact, compelling and convincing

The GUI options at hand are the master of your simulation. Values of the current state of your design are displayed on the screen and you can create a simulation file based on these. Although there is a simulation text control, changes made using the GUI have the upper hand.

Problems encountered pop up on the screen with a brief explanation, setting off a raw console spew to Console text control with which you can easily debug the issue. On successful completion, results are saved in a data file under Simulation Results text control. Results can be observed as graphs, with support from Gwave.

You can also perform a variety of analyses with this tool. Operating point analysis, direct current analysis, alternating current analysis, transient analysis and Fourier analysis are some of the different tests you could use on your design. While transferring from one simulation engine to another, gSpiceUI tries to transfer information between

Promising features at a glance

- Unix-style modular design philosophy
- Choice of two simulation engines
- Consistent UI for different simulation engines
- Choice of two waveform-viewing utilities
- Contents of netlist files displayed and option to edit these
- Schematic capture application that can launch from within the GUI
- Can be compiled on multiple platforms like Linux, FreeBSD, MacOSX and Windows
- Written in C++ using wxWidgets library (no other build dependencies)
- Console display eases fault finding of the toolchain

the files, to reduce the effort from your end.

Prototyping, the design methodology

How does this GUI do what it does? What is the thread that holds its components together and binds these smoothly? Architecture of this application can be explained to a certain extent by five objects this software is modelled based on.

Application Object Model.

Containing the main function, this model depicts the top-level application architecture. Entry point to the application is contained in App class of wxWidgets, extending into that of gSpiceUI.

Main Frame Class Object Model.

This is the top-level GUI architecture that is the source of all display objects inherited from wxWidgets library's wxFrame class.

Analysis Class Object Model.

Inherited from the widget's wxNotebook class, this is the model that depicts the class structure of objects that make the notebook with analysis tabs.

Process Class Object Model.

It is here that the tasks that the GUI can execute are processed. Inherited from wxWidget's wxProcess class, objects that form tasks are structured here.

PnValue Class Object Model.

This model depicts the class structure of the display control that handles integer and floating-point numeric values. It puts together library text, spin and choice controls from wxWidget to let you set values via the mouse or keyboard.

Manoeuvre the easy way

With schematics, circuits, logs and files for various analyses, and different files for the two individual simulation engines, the number of files generated or created can be massive. To help you weave through these easily, the naming system of temporary files generated is thoughtfully created, with files specifics included.

The tool bar provides quick access to most-used functions, and these are grouped together according to activity type to make it easier to work with. On the main window, you can find a list of possible test points, analysis settings, console output and a status bar.

SourceForge holds the primary database for this software, which is licensed under the terms of the GNU general-public licence. wxWidget library is a necessary add-on for compiling the sources. **EFY**

Set your preferences using Preferences dialogue

- Waveform viewer showing different viewer utilities and details of the currently selected one.
- Choice box for temporary file management like deletion or prompting the user. Most of these files contain results and data that might turn out useful even outside the software.
- Choice box containing precision parameters that go in during simulation to round off the concerned data.
- Max-text control, which can be used to set the limit for number of lines in any of the text controls including the results section.
- Sync sweep sources, check box that enables or disables synchronisation of sweep sources.
- Show tool tips for help during design.
- Paths to utilities of the tool; can be set manually.

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